

Segmenting Hippocampus from 7.0 Tesla MR Images by Combining Multiple Atlases and Auto-Context Models

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Abstract. In investigation of neurological diseases, accurate measurement of hippocampus is very important for differentiating inter-subject difference and subtle longitudinal change. Although many automatic segmentation methods have been developed, their performance can be limited by the poor image contrast of hippocampus in the MR images, acquired from either 1.5T or 3.0T scanner. Recently, the emergence of 7.0T scanner sheds new light on the study of hippocampus by providing much higher contrast and resolution. But the automatic segmentation algorithm for 7.0T images still lags behind the development of high-resolution imaging techniques. In this paper, we present a learning-based algorithm for segmenting hippocampi from 7.0T images, by using multi-atlases technique and auto-context models. Specifically, for each atlas (along with other aligned atlases), Auto-Context Model (ACM) is performed to iteratively construct a sequence of classifiers by integrating both image appearance and context features in the local patch. Since there exist plenty of texture information in 7.0T images, more advanced texture features are also extracted and incorporated into the ACM during the training stage. With the use of multiple atlases, multiple sequences of ACM-based classifiers will be trained, respectively in each atlas' space. Thus, in the application stage, a new image will be segmented by first applying the sequence of the learned classifiers of each atlas to it, and then fusing multiple segmentation results from multiple atlases (or multiple sequences of classifiers) by a label-fusion technique. Experimental results on the six 7.0T images with voxel size of $0.35 \times 0.35 \times 0.35mm^3$ show much better results obtained by our method than by the method using only the conventional auto-context model.

1 Introduction

Numerous efforts have been made for developing hippocampus segmentation methods in the field of medical image processing [1-6], since accurate measurement of hippocampus is significant for study of neurological diseases, i.e., Alzheimer's disease. However, due to the tiny size of hippocampus ($\approx 35 \times 15 \times 7mm^3$) and also the complexity of surrounding structures, the accuracy of hippocampus segmentation is limited by the low imaging resolution (often with voxel size of $1 \times 1 \times 1mm^3$).

The learning-based methods [7, 8] have been proven powerful for hippocampus segmentation. Recently, auto-context model (ACM) has been proposed in [9] to automatically segment hippocampus by constructing the classifier based on not only

the image appearance, but also the context features (i.e., a map of classification confidence) in the local patch. In each iteration of ACM, the context information is updated according to the latest classification results and then used to guide the training of the next classifier by Adaboost [10] in the next round of training. To be successful, a large number of image and context features are generally extracted in ACM, which includes intensity, position, and neighborhood features (such as intensity mean and variance, and Haar features at different scales). Although the ACM is effective by integrating the low-level appearance features with the high-level context and implicit shape information, its performance in hippocampus segmentation could still be limited by the distinctiveness of features, which is directly related with imaging resolution.

On the other hand, the high-resolution imaging technique has been rapidly developed after the 3.0T scanner becomes popular in the modern neurological study. It has been shown in [11] that hippocampi can be better manually labeled on 7.0T images than the low-resolution 3.0T images. Thus, researchers have started to apply the previous segmentation algorithms, developed for 1.5T or 3.0T images, to 7.0T images. However, the performance is limited, mainly because the features in 7.0T are much richer and significantly different from those in 1.5T or 3.0T. To demonstrate the rich texture and high enhancement on image contrast in 7.0T image, we show one typical slice with hippocampi (with the resolution of $0.35 \times 0.35 \times 0.35 \text{mm}^3$) in Fig. 1 (b), along with a similar slice scanned by 1.5T scanner (with the resolution of $1 \times 1 \times 1 \text{mm}^3$) in Fig. 1 (a).

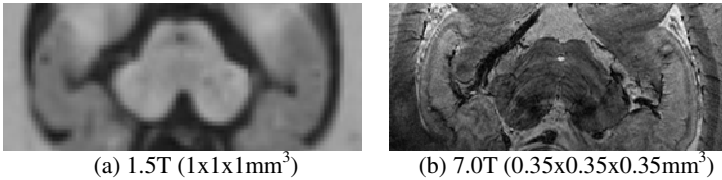


Fig. 1. Large difference between 1.5T (a) and 7.0T (b) MR images

To fully utilize the rich features provided by 7.0T images, we present a new learning-based hippocampus segmentation algorithm, using multi-atlases technique and auto-context models. In particular, ACM is performed to train the classifiers for each atlas and its aligned atlases, based on their respective manually-labeled hippocampi. Importantly, since 7.0T images provide texture information, more advanced texture features are also incorporated into the ACM during the training stage. To segment the hippocampi on a new image, all atlases, along with their respective classifiers learned in the training stage, will be first transformed onto the new image by a registration technique. Then each point in the new image will be classified by the respective classifiers of each atlas, and its final segmentation result will be obtained by fusing all results from all atlases.

Our learning-based hippocampus segmentation method has been evaluated extensively on 7.0T images. It can produce 7%~12% more overlap ratio than the method using only the conventional ACM, because of using multi-atlases technique and also the incorporation of more advanced texture information into the ACM framework in our method.